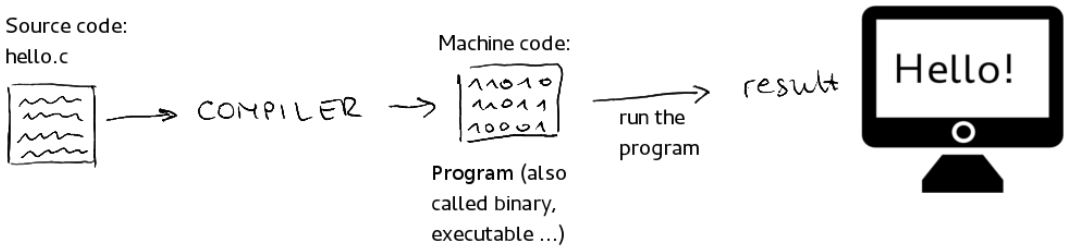
Python is an interpreted programming language

• Many languages require translation or compilation so that the machine can understand it in binary code.



• **Python** is instead **directly interpreted** into machine instructions (at the cost of a bit of computational speed).

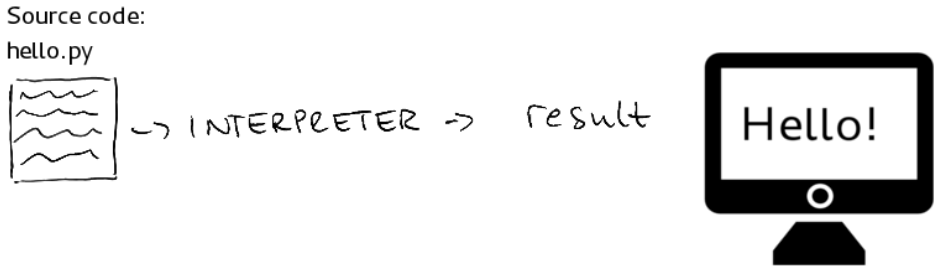
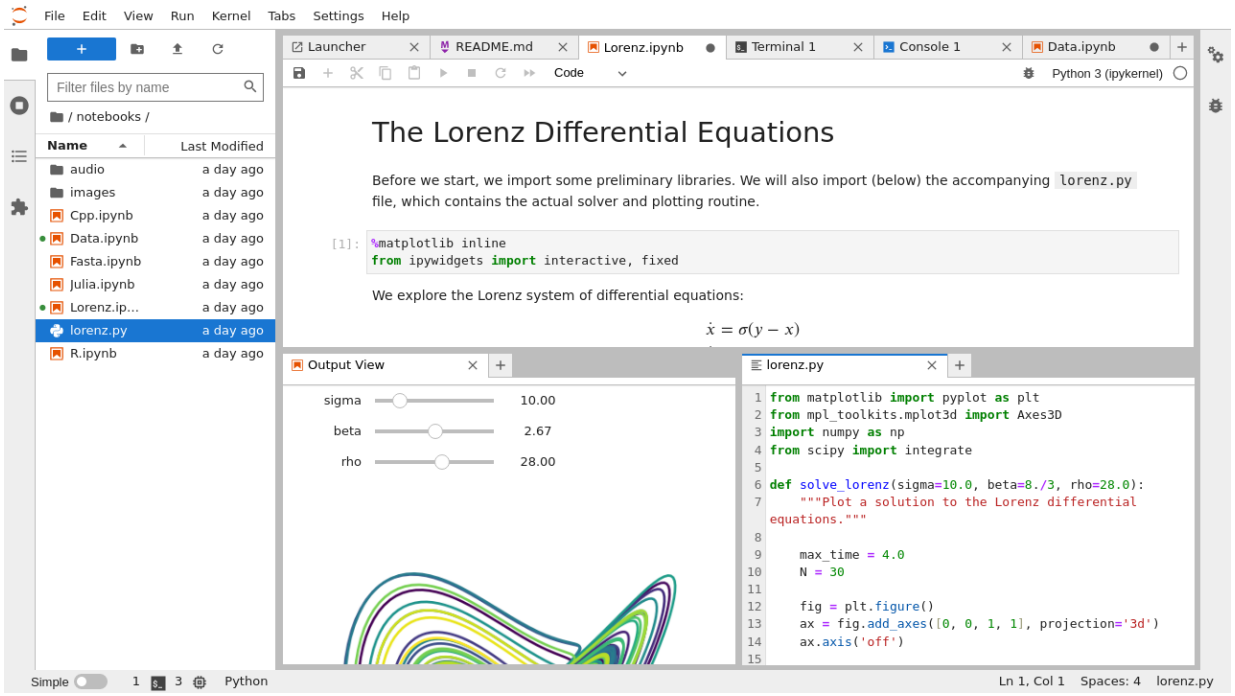


Image credits: https://medium.com/

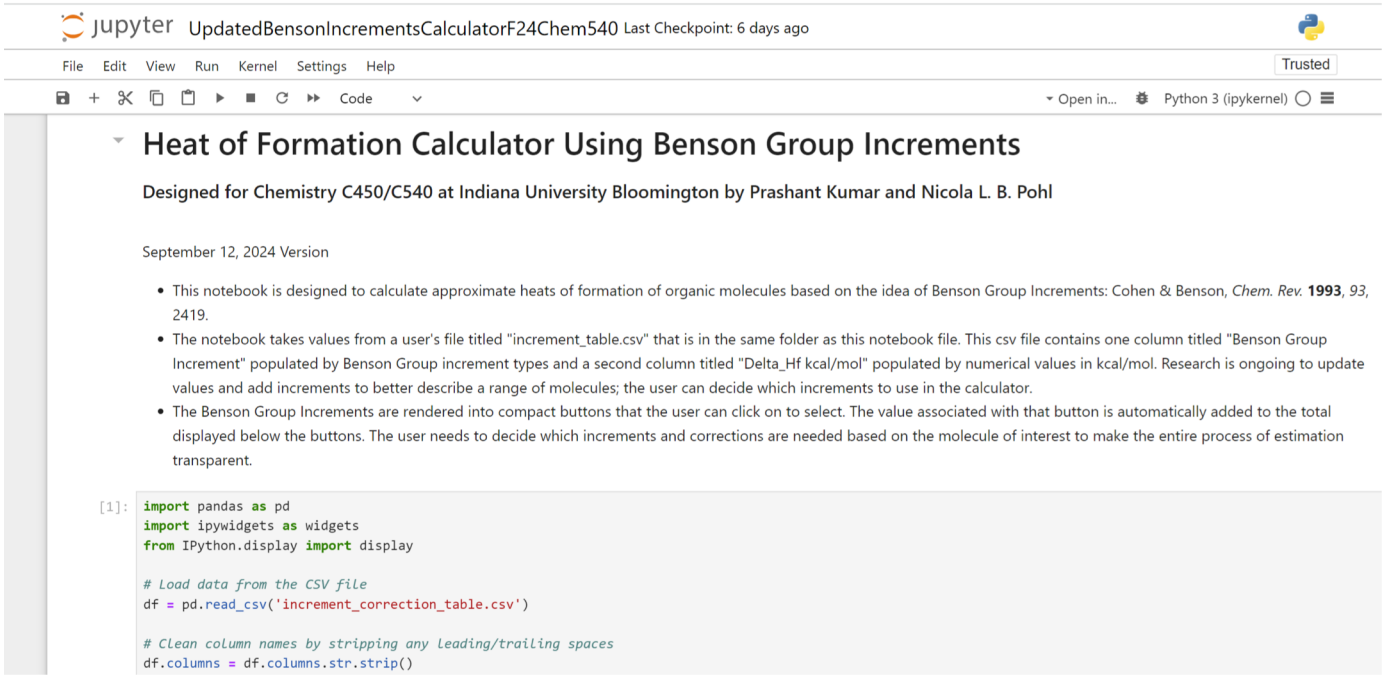
Jupyter Notebook: Integrating Text and Graphics with Code

• Jupyter Notebook is an open-source, free, web-based interactive computational environment for creating lab notebook-like pages that can not only store text and graphics like a paper lab notebook, but *also can run code*.



Jupyter Notebook Is Web-browser Based

- Jupyter Notebook is a browser-based application containing an ordered list of input/output cells which can contain code, text, mathematics, plots and other media.



Installing Python

• Jupyter Notebooks can run code written in Python if that programming language is installed and accessible on your computer.

• There are many ways to install Python on your computer. The most popular methods are:

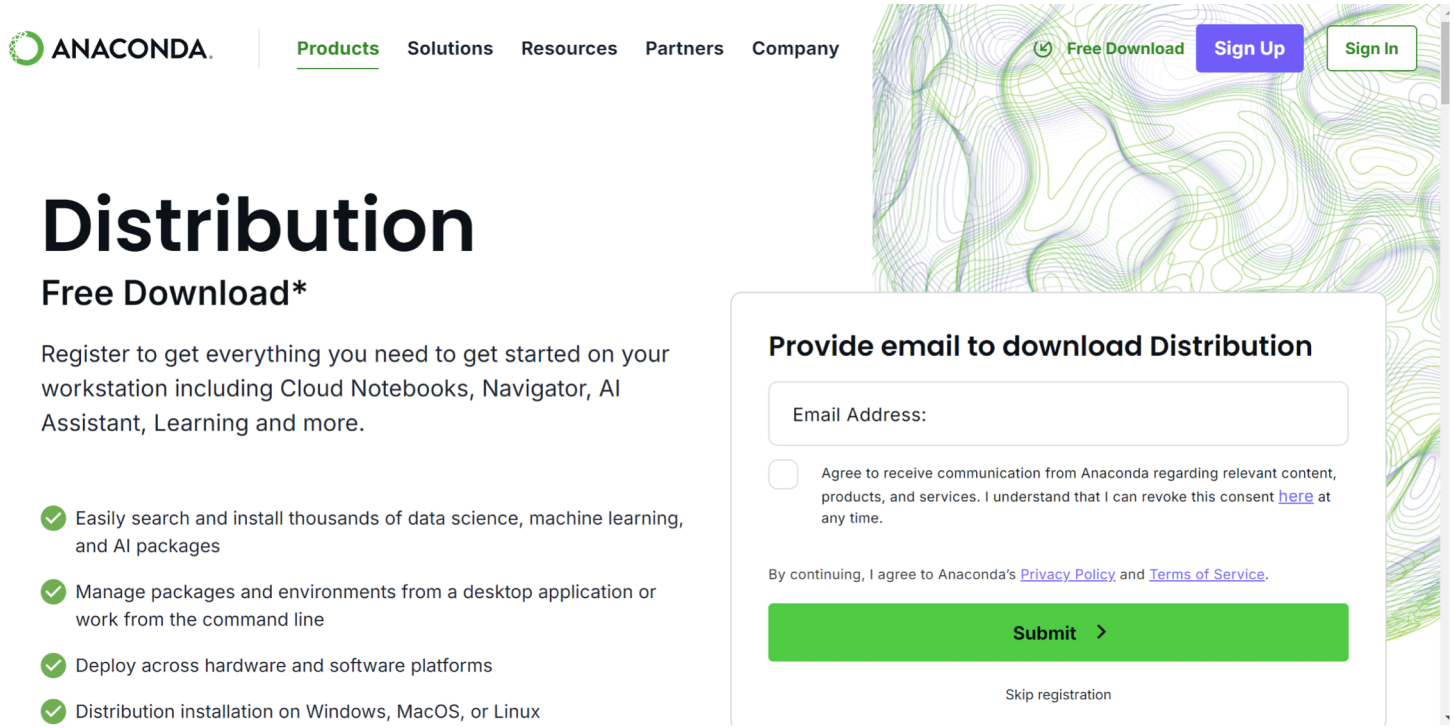
1. Through the official website and use with any source-code editor or web-based application to develop code (for e.g. Jupyter Notebook)

   https://www.python.org/downloads/

2. Easy and preferred option:

   https://www.anaconda.com/download/

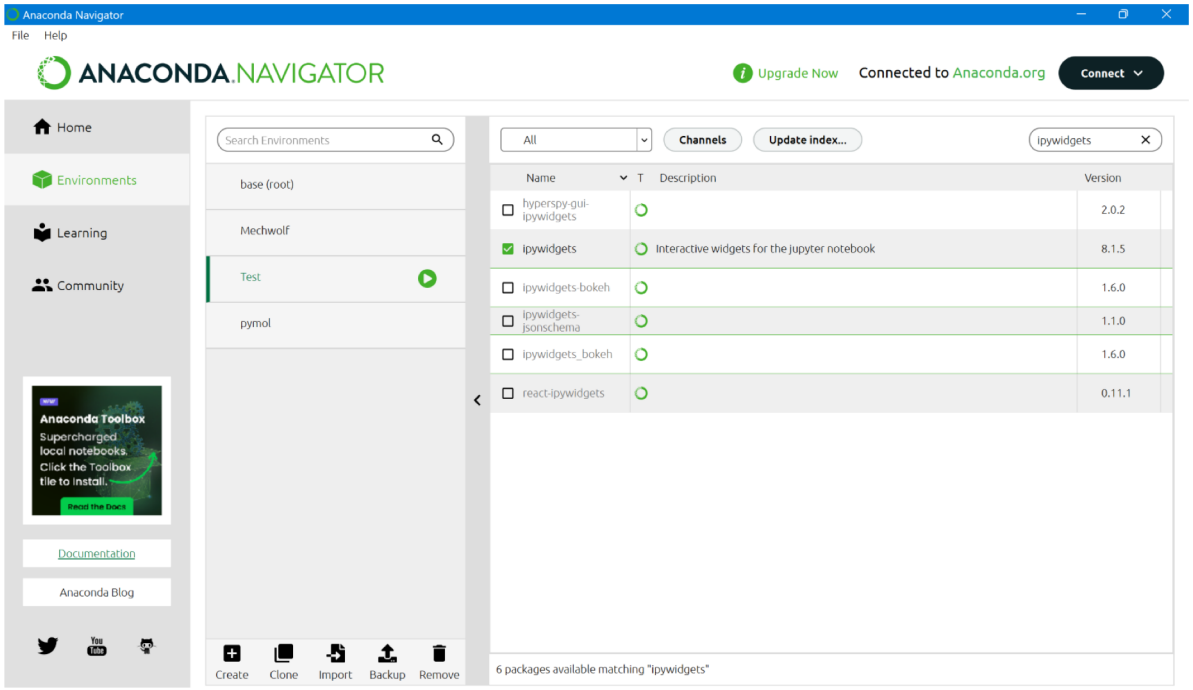
Installing Anaconda App (To Avoid Typing in Terminal Windows)



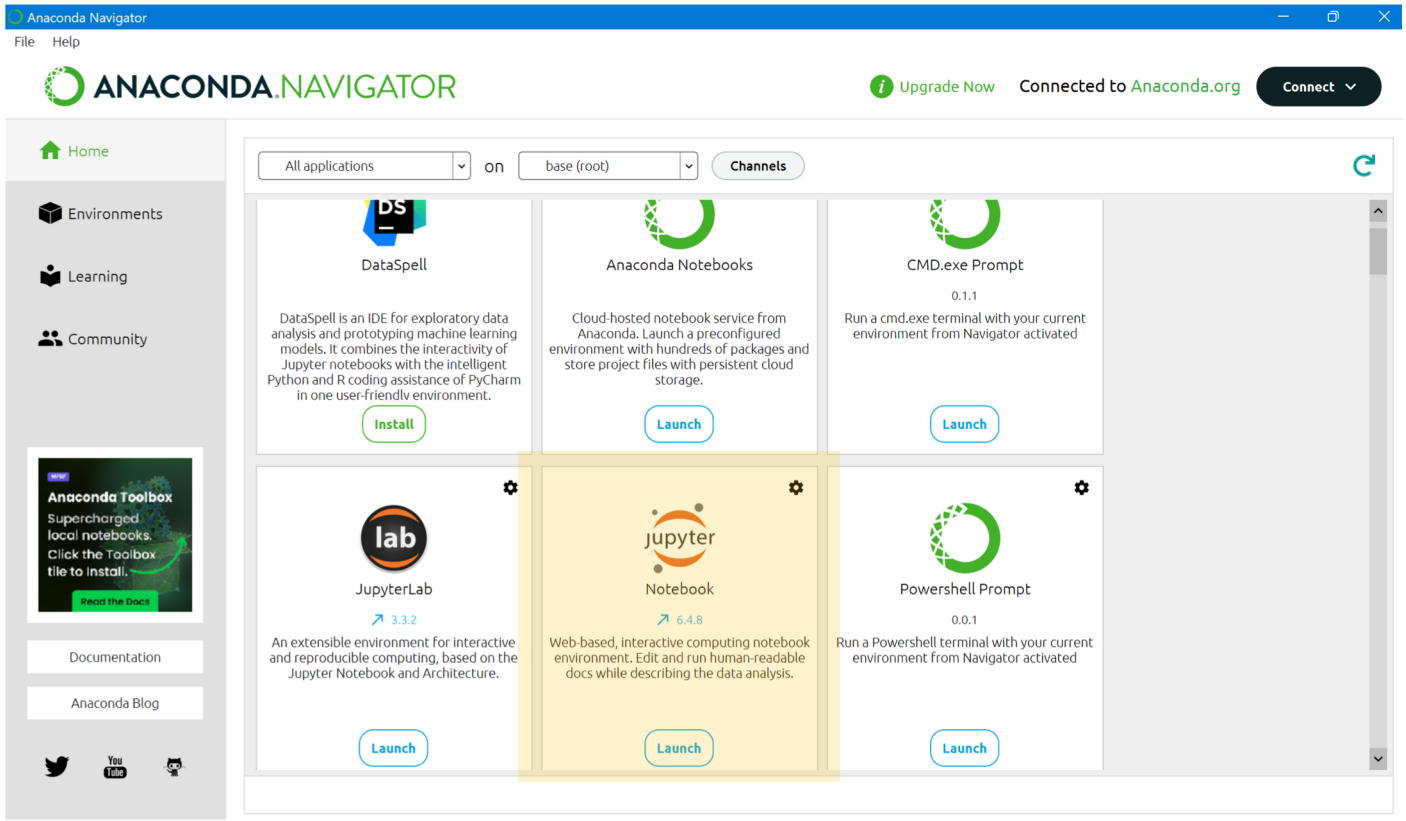
Note: You can skip this step if you already have Python 3.9 with the pandas & ipywidgets libraries installed.

Launch Anaconda and Install Libraries

1. Install Anaconda (~ 5 min) which installs Python
2. Launch Anaconda Navigator
3. Go to Environments. Tap on Create at bottom.
4.Name and create Python 3.9.20.
5. Select All to list all available libraries
6.Install pandas & ipywidgets



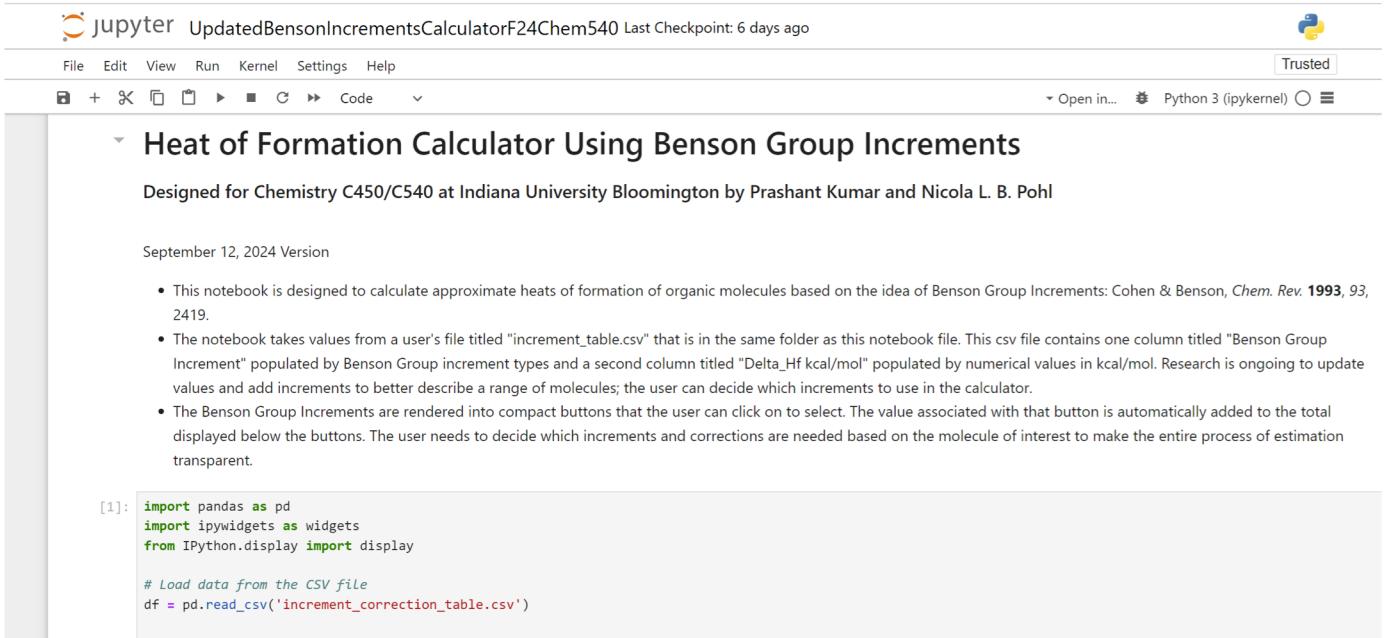
Running Python through Jupyter Notebook



- Return to Home.
- Click on Jupyter Notebook icon to launch a page.

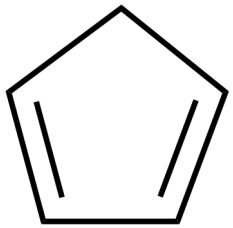
Jupyter Notebook: Benson Increment Calculator

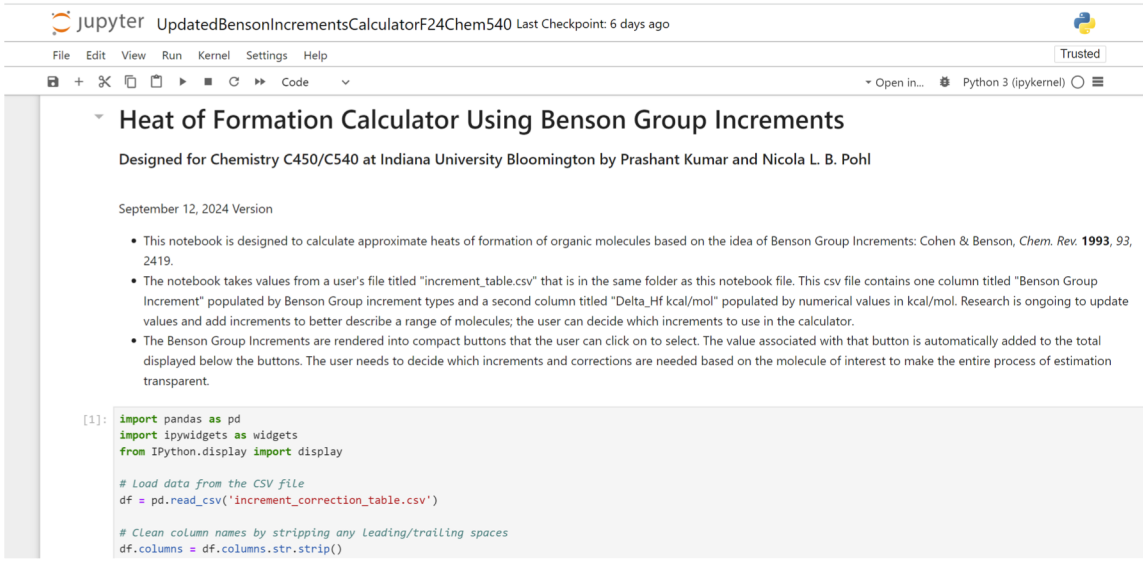
• On Canvas under “Files” in the Jupyter Notebooks/FAIR Chemistry folder, download the Jupyter Notebook page written to easily calculate heats of formation using Benson increments along with the csv file that contains the Benson increment data. Upload this Jupyter Notebook page in your Jupyter Notebook browser window.



Jupyter Notebook: Benson Increment Calculator Problem

• Use the Benson increment calculator to calculate the heat of formation of cyclopentadiene in kcal/mol.





• **Homework:** Look at the code in the Jupyter Notebook. Do you see how you now have a template to easily do other types of calculations just by varying the data file?